CDRF: A Large-Scale RF Benchmark and Toolkit for Drone Detection, Classification, and Open-Set Recognition

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***Abstract*— We present CageDroneRF (CDRF), a large-scale benchmark for Radio-Frequency (RF) drone detection and identification built from real-world captures and systematically generated synthetic variants. CDRF addresses the scarcity and limited diversity of existing RF datasets by coupling extensive raw recordings with a principled augmentation pipeline that (i) precisely controls Signal-to-Noise Ratio (SNR), (ii) injects interfering emitters, and (iii) applies frequency shifts with label-consistent bounding-box transformations for detection. This dataset spans a wide range of new drone models, many unavailable in existing public datasets, and acquisition conditions, all captured exclusively within a controlled RF-cage facility, and is released with interoperable open-source tools for data generation, preprocessing, augmentation, and evaluation that also operate on existing public datasets. CDRF enables standardized benchmarking for classification, open-set recognition, and object detection, supporting rigorous comparisons and reproducible pipelines. By releasing this comprehensive benchmark and tooling, CDRF aims to accelerate progress toward robust, generalizable RF perception models.**

***Index Terms*— Unmanned Aerial Vehicles (UAVs), Radio Frequency (RF), Machine Learning, Deep Learning, Dataset, Spectrograms, Drone Detection, Drone Classification.**

I. Introduction

The rapid proliferation of Unmanned Aerial Vehicles (UAVs) has enabled transformative applications in logistics, agriculture, inspection, emergency response, and transportation [1], while simultaneously introducing acute risks to safety, privacy, and critical infrastructure [2]. High-profile disruptions near airports, increasing reports of unauthorized flights over sensitive facilities, and the use of UAVs in illicit operations underscore the operational need for reliable, scalable counter-UAV systems [3]. Early and effective detection and identification are essential components of any response strategy. [4].

Traditional sensing modalities, radar, vision, and acoustics, exhibit strengths but also notable failure modes: sensitivity to environmental conditions, limited long-range performance for small targets, reliance on line-of-sight or favorable lighting, or high deployment costs. Radio-Frequency (RF) sensing offers complementary advantages. Because most consumer and professional drones maintain continuous RF links for command-and-control and video transmission, RF-based systems can detect and characterize targets day or night, in non-line-of-sight scenarios, and at comparatively low cost. Moreover, RF captures are intrinsically rich for learning, carrying device, protocol, and operator signatures across diverse channels [5].

Despite these advantages, progress in Machine Learning (ML) for RF-based drone perception has lagged behind other domains. A central barrier is the absence of broad, standardized benchmarks and tools. Existing datasets are often narrow in class diversity and limited in raw volume; many are collected in relatively clean environments with narrow Signal-to-Noise Ratio (SNR) ranges and weak interference diversity. As a result, models trained on such data can overfit to idealized conditions, achieving near-perfect scores on "easy" benchmarks yet degrading severely in realistic deployments with Bluetooth/Wi-Fi interference, spectrum crowding, and low SNR. The cost and logistics of collecting sufficiently varied RF captures further slow iteration and reproducibility [6].

We introduce CageDroneRF (CDRF), a benchmark built to close this gap with a dataset-and-toolkit co-design. CDRF comprises real-world captures from controlled RF-cage facilities and open-campus settings, paired with a principled raw-signal augmentation pipeline that program-

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The CDRF dataset, code, and trained models are publicly available at https://github.com/DroneGoHome/U-RAPTOR-PUB.
https://aerodefense.tech/u-raptor-data-request

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matically varies SNR, injects interfering emitters, and applies frequency shifts while preserving label consistency. Crucially, augmentations operate on complex baseband I/Q, before time-frequency conversion, so synthesized conditions faithfully reflect RF phenomena rather than image-level artifacts. For detection, frequency shifts are accompanied by exact recalculation of You-Only-Look-Once (YOLO [7])-format annotations with correct wrap-around behavior on the spectrogram's frequency axis. The data processing stack converts raw captures to spectrograms via Short-Time Fourier Transform (STFT) and exposes all parameters (e.g., sampling rate, FFT size, segment length) for reproducible ablations and reprocessing.

Beyond the dataset, CDRF provides interoperable utilities for dataset creation, cleaning, and evaluation. The data module loads .dat files via memory mapping, slices long recordings into time windows, generates spectrograms, appends per-sample metadata (including time bounds), and optionally adds controlled AWGN, Rician, or Rayleigh fading noise for precise SNR targets. The YOLO toolkit includes raw-IQ augmentation and automatic bounding-box recomputation; a dataset cleaner aligns third-party releases (e.g., Roboflow-style [8]) into a consistent hierarchy; and a lightweight patch exposes per-detection class probabilities from YOLO for richer calibration and analysis. For classification, CDRF ships PyTorch [9] baselines (binary and multi-class) built on ResNet-18 [10] with ready-to-run dataloaders for both spectrogram images and array/pickle formats, standardizing training, metrics, and confusion matrices. All tooling is dataset-agnostic, enabling applications to use existing public datasets to facilitate cross-benchmark comparability.

Collectively, CDRF aims to shift the field from narrow, clean benchmarks toward realistic, stress-tested evaluation under varied SNRs, interference, and frequency offsets, conditions that operational systems routinely encounter. By releasing both data and the underlying signal-first augmentation/evaluation stack, we seek to catalyze reproducible research on RF-based detection, identification, and related perception tasks. We summarize our key contributions as follows:

- **CDRF benchmark dataset:** Real-world RF captures from cage environments with standardized spectrogram generation and rich per-sample metadata.
- **Raw-signal augmentation pipeline:** Controlled SNR injection, interfering-signal mixing, and frequency shifts applied at I/Q level, with exact YOLO label recomputation (including wrap-around).
- **SNR-structured datasets:** Programmatic creation of SNR-stratified splits (including noise-only backgrounds) for stress testing and realistic robustness evaluation.
- **Interoperable tooling:** Open-source utilities for dataset creation, metadata generation, Roboflow-style cleaning, and spectrogram rendering; tools are dataset-agnostic for use on existing releases.
- **Detection analysis enhancements:** Patch exposing per-detection class probabilities from YOLO to support calibration studies and open-set analyses.
- **Baselines and evaluation:** Ready-to-run PyTorch baselines for binary and multi-class classification with standardized metrics and confusion matrices, enabling reproducible benchmarking across tasks.
- **Open set recognition:** Implementation of open-set recognition capabilities to handle previously unseen classes during inference.

The remainder of this paper is organized as follows. Section II reviews related work in RF-based drone detection, covering various sensing modalities and existing public datasets. Section III details the design and collection of the CDRF dataset, and Section IV describes our data preprocessing, augmentation, and annotation pipeline. Section V presents the experimental setup and benchmark results for several machine learning tasks, including drone detection, single-label, open-set, and hierarchical classification. Finally, Section VII concludes the paper and discusses future research directions.

II. Related Work

The landscape of drone detection and classification has evolved rapidly under the pressure of monitoring and mitigating unauthorized UAVs. Research spans multiple sensing modalities, including radar, acoustics, vision, and RF, each with distinct strengths and failure modes. Below, we summarize these modalities, review RF signal representations and learning methods, and discuss public datasets that have shaped the field, highlighting persistent gaps that motivate the development of a new benchmark and tooling.

A. Drone Detection Methodologies

- **Radar-based** [3], [11]–[19] systems exploit micro-Doppler and related motion signatures and can operate at long range and in adverse weather, but often struggle with small, slow, or low Radar Cross-Section (RCS) targets and occlusions.
- **Acoustic-based** [19]–[28] methods rely on propeller/engine signatures and are low-cost and lightweight, yet are highly sensitive to ambient noise and typically limited to short ranges.
- **Vision-based** [26], [29]–[31] systems (RGB/thermal/multispectral) can excel at fine-grained recognition in favorable conditions but degrade with poor lighting, weather, or occlusions, and often require high-resolution optics and line-of-sight.
- **RF-based** [6], [32]–[37] sensing leverages command-and-control and video links that persist during operation, enabling day/night, non-line-of-sight detection with modest hardware costs and rich device/protocol signatures.

- **Multi-Modality** [26], [38]–[43], often referred to as multi-sensor fusion or multi-sensor systems, involves the integration of data from two or more distinct sensing modalities to enhance drone detection and classification capabilities. This approach is seen as a crucial strategy for building truly robust and comprehensive counter-drone systems

To contextualize the role and trade-offs of RF sensing within the broader landscape of drone detection, Table I provides a comparative analysis of major sensor modalities.

B. RF Signal Transformation and Feature Representations

A central challenge in RF-based perception is learning discriminative features from raw I/Q streams. Visual time-frequency representations have become a standard connection to modern machine learning.

- **Spectrograms via STFT** provide localized energy distributions over time and frequency, exposing cues such as center frequency, bandwidth, dwell time, and hop rate that correlate with device and mode. They have proven effective inputs for Convolutional Neural Networks (CNNs) and detectors.
- **Power Spectral Density (PSD)** offers frequency content but discards temporal dynamics; **scalograms** (wavelets [51]) and **Wigner–Ville distributions** [52] trade off resolution, cross-terms, and interpretability.
- **Alternative encodings**, such as Frequency-Domain Gramian Angular Fields (FDGAF) [53], map 1D spectra into 2D images to better preserve amplitude/temporal structure; **wavelet scattering** [54] has also shown promise for bias removal and transient capture on RF benchmarks.

Empirical evidence suggests spectrogram-based models are more robust than raw 1D I/Q pipelines under low SNR and co-channel interference, particularly in Industrial, Scientific, and Medical (ISM) bands congested by Wi-Fi and Bluetooth. [6], [48], [55]

C. Deep Learning for RF-based Drone Detection

Deep learning models dominate modern RF perception pipelines:

- **Deep Neural Networks (DNNs)** [4], [56]–[58] achieved early success for binary detection but often degrade with many classes or confusable spectra.
- **CNNs** [59], [60] on spectrograms or raw I/Q deliver strong accuracy; purpose-built architectures (e.g., CNN-SSDI [61], FDGAF-CNN [55]) report high performance on DroneRF-style tasks.
- **Object detectors** [6], [62], [63] (e.g., YOLO) jointly localize and classify RF emissions on spectrograms, offering center-frequency and bandwidth estimates alongside class labels; lightweight variants target real-time operation.
- **Residual networks** [64] (ResNet-18/50 [10]) serve as strong spectrogram classifiers; more recent Transformers [65] (e.g., Swin [66], ViT [67]), EfficientNet [68], and MobileNet [69] have also been explored for accuracy–efficiency trade-offs.
- **Sequence models** [70] (e.g., Long Short-Term Memory (LSTM) [71], Temporal CNNs [72]) can leverage burst timing; multimodal fusion (RF+acoustics) has shown benefits at low SNR.
- **Classical ML** [73] (e.g., XGBoost [74] on engineered features) remains competitive in binary detection on clean datasets but tends to be less robust under severe interference.

A recurring theme is the sensitivity of performance to data realism: models trained on clean, low-interference datasets show inflated benchmark scores yet generalize poorly under spectrum crowding and low SNR [75], [76].

D. Public Datasets for RF-based Drone Detection

Public datasets have catalyzed progress but reveal a consistent realism gap:

- **DroneRF** [77] provided early stimulus for deep learning with three drone models, multiple modes, and magnitude-only spectra. Collected in controlled environments, it enabled near-ceiling in-benchmark accuracy but limited generalization due to sparse interference and class diversity.
- **DroneDetect / DroneDetect V2** [78] expanded diversity (seven drones, multiple modes) and explicitly included co-channel interference (Wi-Fi, Bluetooth, combined), with complex I/Q captures to support realistic evaluations. However, background-only segments remain limited for training calibrated binary detectors.
- **Cardinal RF** [79] targets classification amid significant interference across drones and non-drone emitters; its restricted availability limits broad benchmarking.
- **VTI_DroneSET_FFT** [80] covers three DJI models across modes with intentional Wi-Fi/Bluetooth clutter (distributed as .mat), emphasizing anti-drone realism while remaining modest in class coverage.
- **Noisy Drone RF Signal Dataset** [81] offers standardized benchmarking across synthetically controlled SNR ranges (e.g., -20 to 30 dB) as preprocessed tensors, facilitating stress tests but limiting raw reprocessing and alternative pipelines.
- **RFUAV** [6] is a large-scale benchmark comprising $\sim$1.3 TB of raw complex I/Q from 37 UAV types, collected in real-world settings with USRPs and accompanied by XML configuration metadata. It provides open preprocessing and standardized evaluation tooling, plus utilities to synthesize controlled SNR conditions from raw captures for robustness

TABLE I: Comparison of Sensor Modalities for Drone Detection

Detection Technology	Primary Advantage (Pros)	Primary Disadvantage (Cons)	Citations
Radio Frequency (RF)	Highly advantageous and cost-effective. Unaffected by weather or lighting conditions. Can operate in Non-Line-of-Sight (NLoS) scenarios. Can detect specific command and control frequencies and offer early warnings.	Extremely sensitive to co-channel interference (e.g., Wi-Fi, Bluetooth). Not perfect for autonomous drones (which may lack continuous RF communication). Limited performance at low SNR.	[2], [44]–[47]
Visual/Optical	Provides high-accuracy identification and is relatively low cost. Often utilizes effective machine learning techniques like Convolutional Neural Networks (CNNs).	Performance is severely degraded by adverse weather and poor lighting. Requires a clear Line-of-Sight (LoS). High false positive rates due to resemblance to birds or other objects.	[3], [36], [46], [48]
Radar	Offers long detection ranges. Less affected by weather and time of day. Provides precise location data.	Conventional radar struggles with mini-drones due to their low Radar Cross-Section (RCS). High expense and regulatory licensing required. Can cause RF interference.	[3], [46], [47]
Acoustic	Cost-effective and passive operation. Can be miniaturized.	Extreme susceptibility to ambient noise. Very short detection range. Cannot detect quieter drones utilizing noise reduction.	[47]–[49]
LiDAR	Offers higher spatial resolution than radar for precise 3D mapping. Excels in low-light or adverse weather.	Requires direct LoS. More expensive and power-intensive than radar. Relatively short range.	[46]
Thermal/Infrared	Can detect heat emitted by objects, performing well in night surveillance and adverse weather.	Prohibitively expensive for long distances. Performs poorly on drones with plastic bodies or electric motors that radiate less heat. May mistake hotter objects for drones.	[49], [50]
Multi-Sensor Fusion	Overcomes the inherent limitations of any single sensor by leveraging complementary strengths. Provides enhanced accuracy and reliability, reducing false alarms.	Increased complexity of system design. Requires advanced sensor fusion techniques to combine data from different formats.	[38], [41], [46]

studies. RFUAV is a public, actively maintained resource designed to prioritize scale, diversity, and reproducibility. Nonetheless, we identify two primary limitations in the current version: (1) its focus on high-SNR recordings relies on synthetic degradations rather than captures of authentic, in-situ interference, and (2) the initial corpus of background-only signals is still limited, with expansion planned through ongoing community contributions.

Despite a growing number of public datasets, their utility is often constrained by persistent challenges, including: (i) a lack of diversity in both device classes and environmental conditions; (ii) restricted access to raw I/Q signals, preventing custom feature extraction; (iii) poorly characterized or narrow SNR distributions; (iv) a scarcity of large-scale, real-world negative samples; and (v) the lack of an integrated framework for augmentation, robust annotation handling, and benchmarked evaluation.

E. Why a New Dataset and Toolkit Are Needed

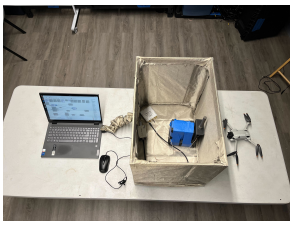
The field is moving from model-centric to data- and system-centric evaluation, demanding benchmarks that: (i) scale in class and environment diversity; (ii) explicitly structure SNR and interference conditions; (iii) release raw I/Q to enable reproducible reprocessing; and (iv) ship *tooling* that enforces label-consistent transforms and standardized evaluation. Prior work seldom provides raw-signal augmentation pipelines that (a) inject controlled noise at baseband, (b) mix interfering emitters realistically, and (c) apply frequency shifts with *exact* annotation recomputation on spectrograms (including wrap-around). Similarly, few detection toolchains expose per-detection class probability vectors for calibration and open-set analyses. Addressing these gaps is essential to move beyond easy benchmark performance toward reliable, fieldable RF perception.

III. CDRF: Design and Collection

In the CDRF dataset, we aim to cover the RF signals emitted from the most common UAV-related devices in the market, including the downlink video signals and uplink Remote Controller (RC) signals. We start building the dataset by recording the UAV signals in a rather clean RF environment, in order to emphasize the characteristics of the signals of interest, and also relieve the impact of interfering RF sources or environmental variations. The clean data can also help on reducing the efforts on data-preprocessing and any following data augmentation processes.

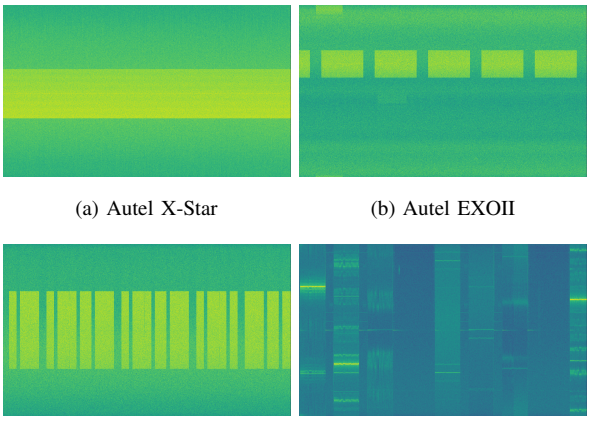


(a) Recording configuration.



(b) Recording equipments.

Fig. 1: Equipment used to capture the data: portable RF shielded enclosure, SDR Card, a laptop, and a drone.



(a) Autel X-Star (b) Autel EXOII

(c) DJI Mavic 2 Pro (d) RadioMaster TX16S

Fig. 2: Representative spectrograms from the CDRF dataset, illustrating the diversity of signal types and device behaviors captured during controlled data collection.

A. Data Acquisition Platform

We built the data collection platform using a Software-Defined Radio (SDR) based system. In order to create an RF-isolation platform, we built a customized Faraday cage and placed both the SDR card and receiving antenna inside the cage to mitigate the interfering RF signals. The hardware setups and platform configurations are shown in Fig. 1.

We use a Lenovo IdeaPad Flex 5 as the data sink. The machine is equipped with an 8-core Intel Core i7-1165G7 CPU and 16 GB of memory. The operating system running on the machine is Ubuntu 24.04.2 LTS. The data collection flow and scripts are built through GNU Radio v3.10 to bridge the Universal Software Radio Peripheral (USRP) B200-mini for RF signal receiving and processing. The SDR device is then connected to a dual-band omnidirectional antenna, which is attached to a customized mount unit to ensure the antenna is static during data collection.

To ensure comprehensive coverage of drone transmissions, we employ sweep scanning across relevant frequency bands during data collection. This approach allows us to detect and record signals from devices operating at variable frequencies.

B. Data Collection Methodology

To initiate the data collection, we set up the platform as described in Fig. 1 (a), then turn on the UAV and the binding controller. While leaving the antenna inside the cage, we place the UAV outside and at a very close distance to the antenna. This placement ensures that the UAV signals delivered to the antenna are strong enough, while other inferencing RF sources in the surrounding areas are at further distances, and thus their signals will be mostly blocked by the Faraday cage. In addition, the RC of the UAV is kept in another room next to the one where the data is collected. Our goal is to minimize the RC signals captured by the collection platform in this phase, so that the dataset can support further analysis on UAV signals and RC signals separately. This can also provide more flexibility for further data augmentation.

Next, we adjust the parameters on both the SDR devices and the UAVs to ensure they are properly aligned for optimal data recording quality. This includes settings such as the gain of the SDR device, the center frequency, and the bandwidth of the UAV's video transmission. In this work, we set the gain of the SDR device to 50 dB for indoor use and 76 dB for outdoor use, with a sampling rate of 20 MHz for both environments. For the video signal from UAVs, we select one of the channels in each of the frequency bands supported by the UAV, e.g., 900MHz, 2.4GHz, or 5.8GHz. Generally, we choose the channels that are not commonly occupied by other non-UAV wireless traffic in order to ensure the quality of the collected data. After the UAV transmission channel is defined, we tune the center frequency of the SDR device to align it to the UAV channel. This ensures that any UAV signals with a bandwidth less than or equal to 20MHz can be fully captured. In addition, this frequency alignment can help simplify the data preprocessing step for any research focusing on frequency domain representation of the signals, because the resulting transformation from the raw data, e.g., spectrogram, will contain a centralized UAV signal.

After that, we run the data recording script from the laptop for a predefined time duration, and the raw data is generated and stored in a `{drone_manufacturer}_{drone_model}_{bandwidth}_{drone_center_freq}_{drone_operation_mode}.dat` file along with all the predefined labels. In CDRF, we record the UAV manufacturer, model, center frequency of UAV signals, and bandwidth for each recording in the corresponding filename or directory name.

Finally, we implement additional software scripts, including a signal playback module in GNU Radio, to load the raw data file, to review the validity of the recorded raw data, and to ensure the signals recorded match the predefined parameters. We also estimate the SNR and the channel condition to make sure the recorded signals are

in a rather clean state. Fig. 2 shows some representative spectrograms from the CDRF dataset, illustrating device behaviors captured during controlled data collection.

C. Collection of Outdoor Data

To complement our controlled indoor recordings, we conducted extensive outdoor data collection sessions. This effort was aimed at enhancing the diversity and realism of the CDRF dataset by capturing UAV signals in complex RF environments, which better reflect real-world operational conditions with various signal sources and environmental factors.

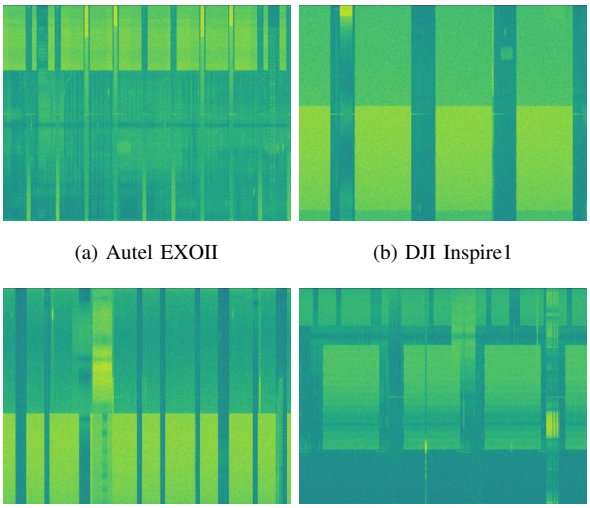
For each outdoor session, we meticulously recorded a comprehensive set of metadata to ensure the data's utility and reproducibility. The raw I/Q data files are organized into a structured directory hierarchy where each directory name systematically encodes the key parameters of the recording session. The naming convention is as follows:

`{device}_{status}_{env}_{sdr_gain}_{splitter}_{duration_recording}_{distance}_{altitude}_{center_freq}_{drone_c_freq}_{bw}_{snr}_{sampling_rate}_{record_dir}.dat`

The fields in the directory name are defined as:

- `device`: The model of the drone or device being recorded.
- `status`: The operational state of the drone (e.g., hovering, flying, on the ground).
- `env`: The environment where the recording took place.
- `sdr_gain`: The receiver gain of the SDR in dB.
- `splitter`: A flag indicating if a signal splitter was used.
- `duration_recording`: The total duration of the recording in seconds.
- `distance`: The horizontal distance from the receiver to the drone in meters.
- `altitude`: The altitude of the drone in meters.
- `center_freq`: The center frequency of the SDR receiver in MHz.
- `drone_c_freq`: The drone's transmission center frequency in MHz.
- `bw`: The signal bandwidth in MHz.
- `snr`: The estimated SNR in dB.
- `sampling_rate`: The sampling rate of the SDR in MHz.
- `record_dir`: The name of the directory where the recording is stored.

Fig. 3 presents a selection of representative spectrograms from our outdoor data collection, featuring signals from various drones such as the Autel EXOII, DJI Inspire1, DJI Mavic3, and DJI Phantom3 Advanced. In stark contrast to the clean, isolated signals captured indoors (as seen in Fig. 2), these outdoor spectrograms exhibit significantly higher levels of background noise and interference. This is visually evident from the brighter and more cluttered appearance of the spectrogram backgrounds, a direct result of the complex real-world RF environments in which the data was collected. The inclusion of this noisier, more challenging data is crucial for developing and validating drone detection models that are robust enough for practical, real-world deployment.



(a) Autel EXOII (b) DJI Inspire1

(c) DJI Mavic3 (d) DJI Phantom3 Advanced

Fig. 3: Representative spectrograms from the outdoor dataset, showcasing the variability in signal characteristics due to environmental factors.

D. Dataset Composition and Signal Characteristics

The CDRF dataset is designed to provide a comprehensive and diverse collection of RF signals for drone detection and identification research. The dataset is built upon a large corpus of raw signal captures, totaling approximately 500+ GB of raw data.

At the core of our dataset is a significant diversity of drone models and signal types. The collection includes signals from 39 unique classes, encompassing 23 distinct commercial and hobbyist UAV models, their remote controllers, and operational variants (e.g., armed vs. unarmed states). This variety, spanning major manufacturers such as DJI, Autel, and Parrot, ensures that models trained on CDRF are exposed to a wide range of communication protocols, a substantial leap from the handful of models found in many previous datasets. In addition to drone signals, we include a substantial number of "no-drone" recordings, capturing ambient RF noise and other common interferers like Wi-Fi signals. This enables the training of robust binary classifiers that can distinguish between the presence and absence of drone activity, a critical requirement for real-world deployment. To further challenge detection and identification algorithms, the dataset also contains recordings of multiple drones operating simultaneously in the same airspace.

All raw data is stored using the In-phase and Quadrature (I/Q) sampling method. Each sample is represented as an interleaved binary bitstream of two 32-bit floating-point numbers, capturing the full complex signal. This raw format is crucial as it allows researchers to develop and test novel signal processing and feature extraction techniques that operate directly on the baseband signal, prior to any transformations like the STFT.

The signals captured in CDRF exhibit a variety of characteristics typical of modern drone communication systems. Many of the drones utilize Frequency-Hopping Spread Spectrum (FHSS) for their command and control links. These signals are defined by key parameters such as their hopping frequency, duration, duty cycle, and the period of their hopping pattern. In contrast, signals used for video transmission typically have wider bandwidth and longer duration than control signals. We define the collection of these characteristics as the RF Drone Fingerprint, which serves as a rich input for deep learning models to differentiate between drone types.

Furthermore, our data captures the inherent behavioral variability of drone signals. For instance, the frequency hopping patterns of a drone during its initial pairing process can differ significantly from those during normal flight operations. The dataset also captures more subtle phenomena, such as the variable time duration of video transmission signals from certain drone models, a characteristic that can serve as another important feature for robust classification.

IV. Data Preprocessing, Augmentation, and Annotation

We release both a raw-signal processing pipeline and augmentation utilities designed to create realistic, label-consistent training data from complex I/Q captures. All components are open-source and parameterized for reproducibility.

A. Spectrogram Generation

Raw complex I/Q streams are transformed into time–frequency representations via the STFT. Given a complex signal $x[n]$, window $w[n]$, window length N, and hop size H, the STFT is [82]

$$X(m,k) \;=\; \sum_{n=0}^{N-1} x[n+mH]\; w[n]\; e^{-j2\pi kn/N},$$

where m is the discrete time index and k is the discrete frequency index.

We compute the power spectrogram $S(m,k) = |X(m,k)|^2$ and render in dB as

$$S_{\mathrm{dB}}(m,k) \;=\; 10\log_{10}\big(S(m,k)+\epsilon\big),$$

where ϵ represents a very small, positive constant used to avoid taking the logarithm of zero. After that map to RGB using a perceptual colormap. By default, we use `scipy.signal.stft` [83] with a Hann window, $N =$ `FFT_SIZE` $= 1024$, overlap $= 128$ samples, and return two-sided spectra with DC centered via `fftshift`. Default sampling is `SAMPLING_RATE` $= 20$ MHz and we generate `NUM_FFT_SPEC` $= 1500$ time bins per sample. Spectrograms are exported as PNG images alongside per-sample metadata.

B. Frequency Resolution and Colormap Selection

The window size N and hop $(N - \text{overlap})$ set the time–frequency trade-off; larger N improves frequency resolution at the cost of temporal precision. All parameters are exposed to support ablations. We provide multiple colormaps (`viridis`, `plasma`, `inferno`, `magma`, `cividis`, `gray`, `hot`) to test color sensitivity; the default pipeline normalizes spectrograms to $[0,1]$ before colorization and composites to RGB for consistent rendering.

C. Data Normalization

We include signal-level and spectrogram-level normalization operators:

- *Power normalization* of complex I/Q to unit average power: $\tilde{x} = x/\sqrt{\mathbb{E}|x|^2+\epsilon}$.
- *Z-normalization* of real and imaginary parts independently to zero mean, unit variance.
- *Spectrogram normalization* options: per-frequency z-score, per-time z-score, or global min–max to $[0,1]$ prior to colorization.

These are selectable in the augmentation scripts and can be toggled per experiment.

D. Custom Augmentation Tools

Our augmentation operates at the *raw-signal* (I/Q) level to better preserve RF physics and then projects to spectrograms, with exact label propagation to detection annotations.

1. *SNR Conditioning via AWGN*

To synthesize controlled SNRs, we add complex AWGN $n \sim \mathcal{CN}(0,\sigma^2)$ to x where the noise power σ^2 is chosen for target $\mathrm{SNR}_{\mathrm{dB}}$:

$$P_s = \mathbb{E}|x|^2, \quad \mathrm{SNR}_{\mathrm{lin}} = 10^{\mathrm{SNR}_{\mathrm{dB}}/10}, \quad \sigma^2 = P_s/\mathrm{SNR}_{\mathrm{lin}}.$$

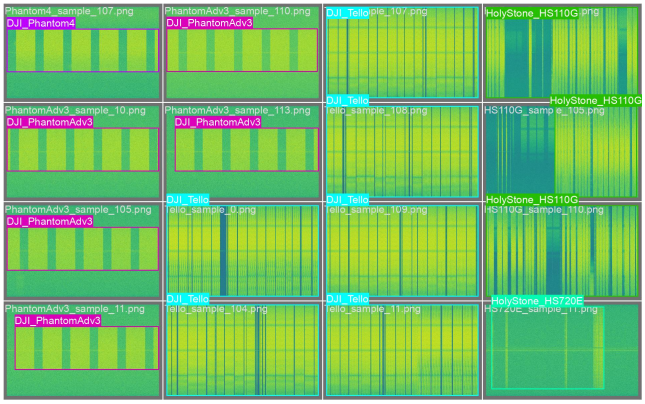
We optionally export noise-only spectrograms for analysis and create SNR-stratified dataset variants (e.g., per-SNR subdirectories), enabling robustness sweeps.

2. *Frequency Shifting*

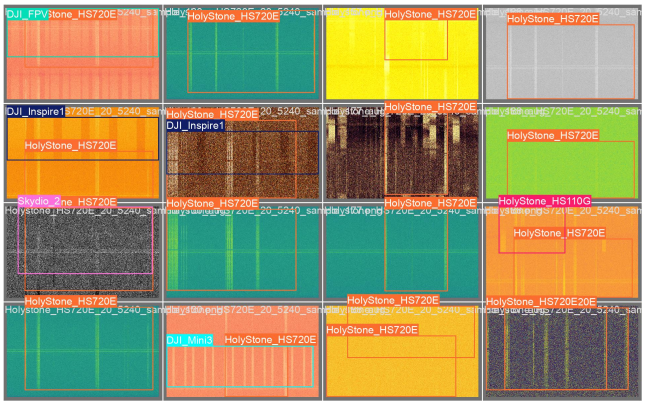
We simulate carrier offsets and front-end errors by multiplying with a complex exponential

$$x_{\Delta f}[n] \;=\; x[n]\; e^{j2\pi\Delta f\, n/F_s},$$

which induces a vertical translation on the spectrogram and where F_s is sampling rate. This is applied before STFT and is label-consistent with our detection annotation recomputation (below).



(a) Annotated Data.



(b) Augmented Data.

Fig. 4: Examples of annotated and augmented data.

3. Interferer Mixing

To emulate spectrum crowding, we mix two normalized signals at a controllable ratio $\alpha \in [0, 1]$:

$$x_{\text{mix}} = \text{norm}\left(x_1 + \alpha\, x_2\right),$$

optionally with independent frequency shifts on each source. This produces realistic co-channel scenarios directly in the I/Q domain.

4. Spectrogram Rendering Variants

Augmented samples are rendered under randomized, but reproducible, choices of colormap and normalization policy to increase visual diversity while keeping frequency content intact.

5. Background Noise and Non-Target Signals

Our dataset includes a comprehensive corpus of non-target signals, captured from diverse environments where no drones were active. To facilitate common machine learning workflows, we provide a suite of tools to extract these "no-drone" segments and generate unified metadata files for mixed-signal experiments. This curated collection of real-world negative samples is crucial for training well-calibrated binary detectors and is explicitly designed to support advanced strategies like hard-negative mining.

E. Annotation Strategy

We adopt YOLO-format annotations on spectrogram images with a *whole-signal* policy: contiguous RF emission bursts from a device are annotated as a single object to capture both spectral morphology and temporal occupancy. Fig. 4 shows examples of annotated data and augmented data with annotations.

In general, a spectrogram of a UAV signal segment consists of a sequence of ON-OFF states, which represents the transmission pattern of the UAV in both time and frequency domain. In order to lead YOLO model to better capture the signal characteristics in both domains, we define the annotation policy as that, the bounding box for each type of signals in a spectogram image will start and end with an ON state of the signal. The OFF states at the beginning and end of a spectrogram are neglected. This can help the YOLO model to track the start and end of a transmission, and thus to output more accurate bounding boxes. Meanwhile, if the RF emission bursts captured by a spectrogram are with limited time duration, i.e., $< 10\%$ of the time duration of the spectrogram, or there is only a single ON state in the whole spectrogram, no bounding box annotation will be given, and the spectrogram image will be labeled as background in those cases.

As for the RC signals, due to its frequency-hopping property, in CDRF we aim at providing two types of YOLO-formated annotation to support the flexibility of future research or implementation in potentially different purposes.

- We provide one bounding box covering all the RC signal bursts captured in a spectrogram. Similar to the UAV signal labeling, we ensure that the edges of the bound box align with the start and end of the sequence of RC signal bursts
- CDRF also contains annotation for RC signal bursts transmitted at the same channel. Since the frequency-hopping behavior is not guaranteed to occur or being captured in every single spectrogram in a continuous raw data recording, we believe that this type of annotation can contribute the learning of YOLO models and improve the detection performance when deploying it in reality.

Moreover, due to the fact that most of the RC signals are narrowband and in short duration for each ON state, we do not provide any bounding box annotation to the spectrogram that contains only one single ON state. The reason behind this design choice for annotation strategy is that the bound box covering only single ON state does not provide any information for the model to learing about its pattern across either time or frequency domain. Similar statement on UAV video signal annotation is also demonstrated in [84]. Also, the detection or classification based on a single ON state is generally neither expected nor considered reliable in real-world UAV detector deployment scenarios.

1. Label-Preserving Transforms with Wrap-Around

Frequency shifts correspond to vertical translations in normalized image coordinates by $\Delta y = \Delta f / F_s$. We recompute bounding boxes analytically after augmentation and correctly handle wrap-around at the top/bottom edges: if a box straddles the boundary after shift, it is split into two valid boxes whose heights sum to the original. Boxes are also filtered by a configurable minimum height to avoid degenerate labels after extreme shifts.

2. Dataset Hygiene and Third-Party Integration

We include utilities to normalize third-party releases (e.g., Roboflow-style layouts) into a consistent hierarchy (`images/ labels/` with class folders and {train,val,test} splits) and to preserve class mappings. This enables direct augmentation and training across heterogeneous sources.

F. Reproducibility and Metadata

Each sample carries rich metadata to ensure reproducibility: original file path, device/model label, sampling rate, FFT size, number of STFT frames, per-sample time bounds (start/end in seconds), center frequency, and output paths for spectrograms. All processing parameters (e.g., window size, overlap, SNR target, shift amounts, mixing weights) are saved or deterministically re-creatable from configuration, facilitating exact regeneration of training splits and ablation studies.

V. Machine Learning Tasks and Baselines

This section details the machine learning tasks supported by our dataset, the baseline models implemented, and their performance. We address drone detection, single-label classification, multi-label classification, open-set recognition, and hierarchical classification.

A. Drone Detection (YOLO)

1. Task Definition

Given a time–frequency spectrogram, the goal is to detect and localize RF emissions attributable to drones. The detector predicts a set of axis-aligned bounding boxes with class labels in YOLO normalized coordinates, where each object corresponds to a contiguous emission burst (whole-signal policy) on the spectrogram.

2. Baseline Model Architecture and Training

We use a single-stage object detector, `YOLOv11n` [85] (Ultralytics [86] v8.3.156), trained on spectrograms produced by our pipeline (section IV). We evaluate two scenarios: (i) Clean, in which spectrograms are generated without augmentation; and (ii) Augmented, in which raw complex I/Q are transformed before STFT via frequency shifting, interferer mixing, and SNR conditioning. Models are initialized from COCO-pretrained weights [85]–[87], trained with deterministic seeds, and validated at each epoch. Labels follow the whole-signal policy, and for frequency shifts we recompute YOLO annotations exactly with wrap-around handling on the frequency axis to preserve label consistency. Unless otherwise specified, Ultralytics defaults are used for input size, data augmentation, and non-max suppression.

3. Evaluation Metrics

We report standard detection metrics: class-aggregated Precision and Recall; mean Average Precision at IoU 0.5 (mAP@0.5); and COCO-style mAP averaged over IoU thresholds from 0.5 to 0.95 in steps of 0.05 (mAP@[.5:.95]). For qualitative error analysis, we also include normalized confusion matrices computed from matched detections.

4. Experimental Results and Baseline Performance

TABLE II: YOLO detection results.

Run	Epochs	Precision	Recall	mAP@0.5	mAP@[.5:.95]
Clean	100	0.932	0.982	0.986	0.948
Augmented	50	0.940	0.855	0.935	0.838

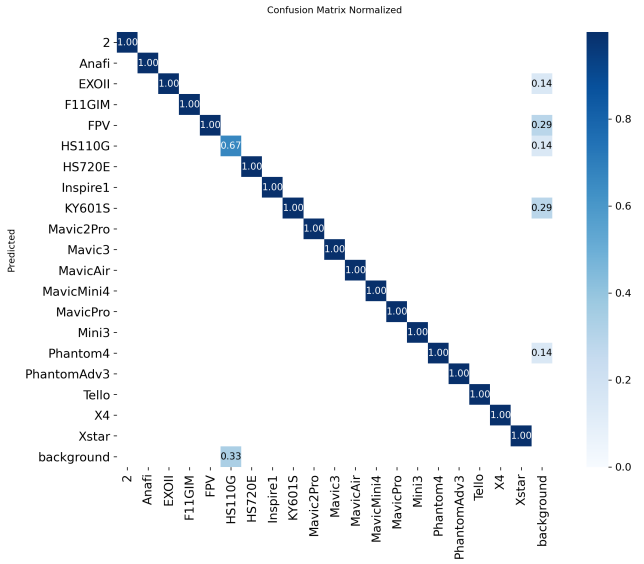
Table II summarizes the best validation checkpoints: The model trained without any augmentations (referred to as "clean") achieves strong performance (Precision $=$ 0.932, Recall $=$ 0.982, mAP@0.5 $=$ 0.986, mAP@[.5:.95] $=$ 0.948), with the top mAP@[.5:.95] observed near epoch 78. Under augmentation, precision is maintained and slightly improved (Precision $=$ 0.940) while recall drops (Recall $=$ 0.855), yielding lower top-line mAP (mAP@0.5 $=$ 0.935, mAP@[.5:.95] $=$ 0.838; best near epoch 49). This reflects the expected precision–recall trade-off when training on harder, low-SNR and interferer-rich conditions: the detector becomes more conservative (fewer false positives) at the expense of increased misses (more false negatives).

Despite this shift, the normalized confusion matrices in Fig. 5 remain strongly diagonal for both regimes, indicating that whole-signal annotations preserve class separability even under frequency shifts and spectrum crowding. The dominant degradation under augmentation is recall (missed detections) rather than systematic label confusion, suggesting room for robustness gains via longer training, class-balanced sampling, or stronger backbones without altering the annotation policy.

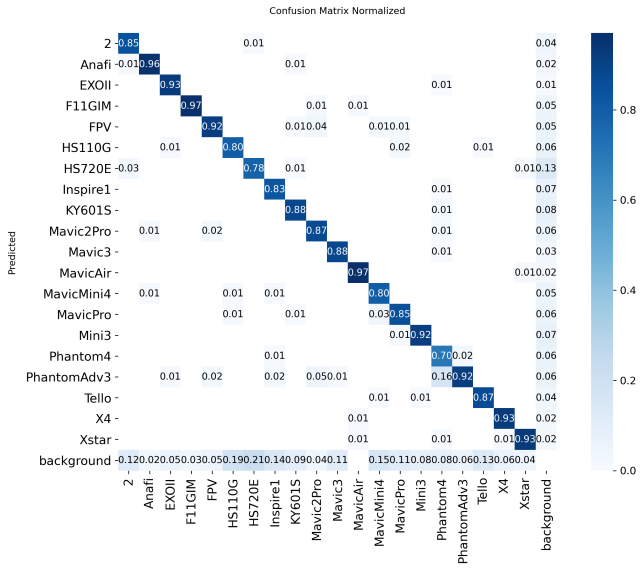
B. Hierarchical Classification

1. Task Definition

To provide more robust and interpretable classifications, we explored a hierarchical approach that classifies signals in a coarse-to-fine manner. In Fig. 6, we provide the full structural tree of the hierarchical classification. Our hierarchy is structured into three levels: Modulation (3 classes), Protocol (5 classes), and finally the specific drone Model (27 classes).



(a) Clean: normalized confusion.



(b) Augmented: normalized confusion.

Fig. 5: Class-wise performance and error structure on validation.

2. Baseline Model Architecture and Training

We implemented a multi-task learning model using a shared ResNet-18 [10] backbone to extract features from the input spectrograms. The output of the feature extractor is fed into three separate classification heads, each dedicated to one level of the hierarchy. The model is trained end-to-end using a "HierarchicalLoss", which is a weighted sum of the cross-entropy losses from each head.

To observe the effectiveness of our indoor data and how it generalizes to outdoor scenarios, we trained our hierarchical classifier on the indoor subset of CDRF and evaluated it on the outdoor subset. Furthermore, we examined how including outdoor data in the training set affects performance by training another model on the combined indoor and outdoor data. In both cases, the amount of indoor training data was kept constant to ensure a fair comparison. The models face challenging scenarios, including varying SNR levels, signal bandwidths, outdoor drones flying at different altitudes, distances, and the presence of various interferers.

3. Experimental Results and Baseline Performance

The results, shown in Fig. 7, compare the generalization performance of two training configurations when tested on outdoor data across three classification levels: Modulation, Protocol, and Model.

When the model is trained on indoor data only, it achieves moderate accuracy on lower-level features (Modulation: 69.95%, Protocol: 66.11%) but struggles significantly with identifying the specific drone model (42.07%). This indicates poor domain generalization, as the model fails to adapt from indoor to outdoor environments.

In contrast, training on a combined set of indoor and outdoor data yields substantial improvements across all hierarchical levels. Accuracy increases to 89.42% for Modulation, 87.34% for Protocol, and 69.63% for Model. The most significant gain of +27.56% is observed in model-level classification, the task most sensitive to environmental shifts. This demonstrates that training on diverse, mixed-domain data is crucial for learning robust, invariant features.

C. Open-Set Recognition

1. Task Definition

Open-Set Recognition (OSR) extends classification to simultaneously: (1) accurately classify known drone types from the training set, and (2) identify and reject unknown samples as "novel" or "outlier" classes, preventing forced misclassification into known categories.

2. Baseline Model Architecture and Training

In our experiments, we utilized MetaMax [88], a state-of-the-art OSR method that combines deep feature learning with statistical modeling of known class distributions. For our OSR experiments, we curated a subset of CDRF comprising 14 distinct drone types totaling 2,180 spectrogram images. The dataset was strategically partitioned into known and unknown classes to simulate realistic deployment scenarios. The known class set contains 10 drone types with 1,915 images, and the unknown class set comprises 4 drone types with 265 images. The data was split into training (60%) and validation (20%) samples from known classes only, with the remaining samples reserved for testing.

3. Evaluation Metrics

Performance metrics included: (1) closed-set accuracy on known classes, (2) unknown detection rate, and (3) overall accuracy.

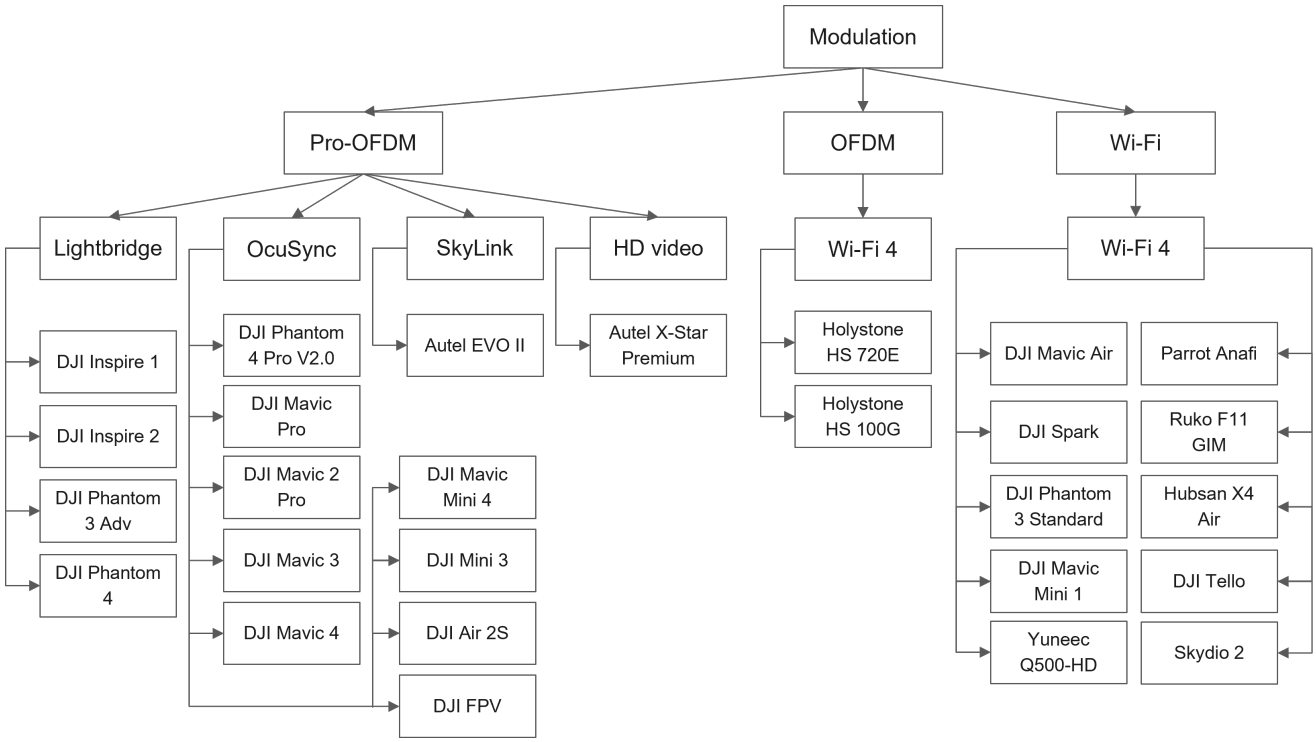


Fig. 6: Hierarchical classification tree structure of CageDroneRF.

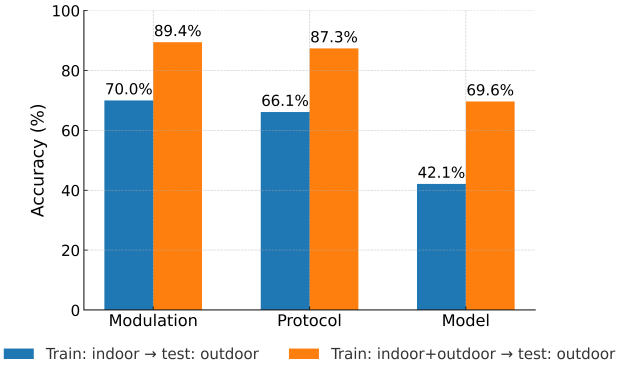


Fig. 7: Hierarchical classification accuracy on different training setups.

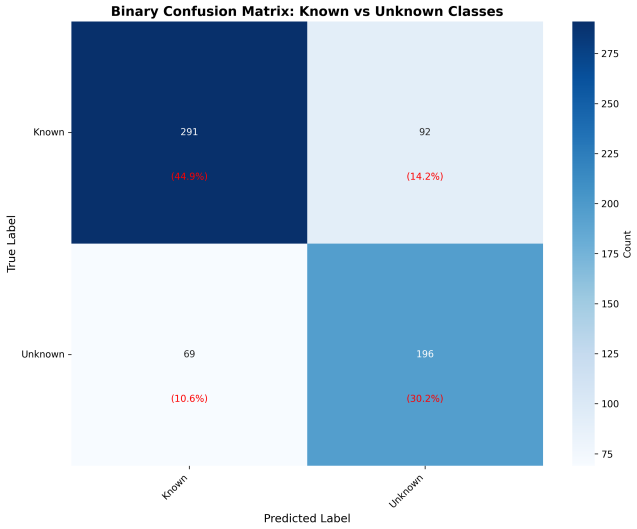


Fig. 8: Binary confusion matrix for known vs unknown class discrimination.

4. Experimental Results and Baseline Performance

We evaluated MetaMax on our curated CDRF subset, achieving an overall accuracy of 75.15%, with known class accuracy of 75.98% and an unknown detection rate of 73.96%. The detailed confusion matrix (Fig. 8) reveals strong performance correlation with class size. Among the 265 unknown test samples, 196 (73.96%) were correctly identified as novel. The binary confusion matrix for known vs. unknown discrimination is shown in Fig. 9. These results demonstrate both the promise and challenges of OSR for RF drone detection.

D. Drone Classification (Single-Label)

1. Task Definition

The single-label drone classification task involves assigning each detected RF signal to exactly one drone type from a predefined set of classes.

2. Baseline Model Architecture and Training

We implemented a ResNet-18 convolutional neural network as the backbone, leveraging its proven effectiveness for spectrogram-based RF signal classification. The

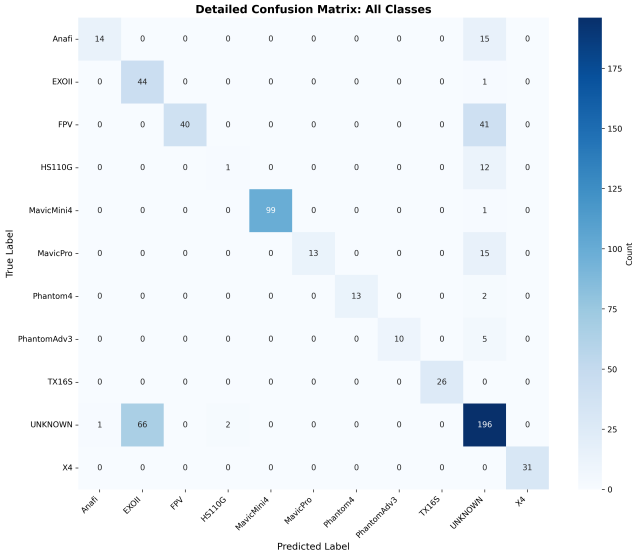


Fig. 9: Detailed confusion matrix showing per-class performance for open-set recognition.

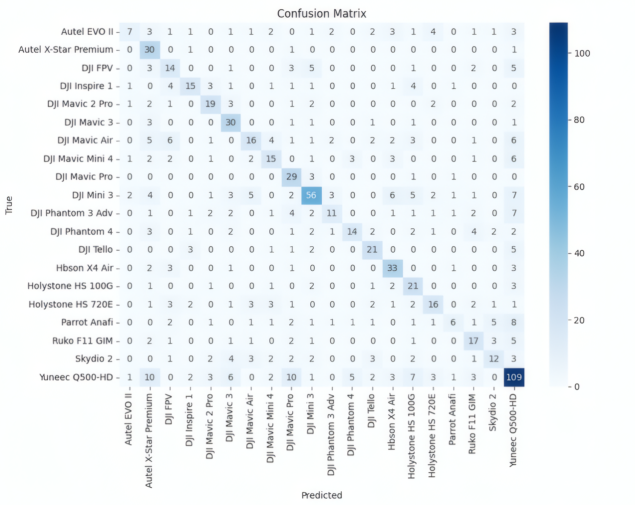


Fig. 10: Confusion matrix for single-label drone classification.

model was trained on spectrogram images generated from the CDRF dataset, using standard data augmentation and normalization techniques to improve generalization. We had a total number of 20 different drone types in our dataset. The ResNet-18 model was trained with a cross-entropy loss and optimized using Adam.

3. *Experimental Results and Baseline Performance*

After training, the model achieved a test accuracy of 53.49%, as shown in the confusion matrix (Fig. 10). The confusion matrix reveals that certain drone types are classified with high accuracy, while others exhibit significant misclassifications, especially for classes that seen less training data, indicating areas for potential improvement. The results highlight the challenges of single-label classification in RF drone signals, particularly in distinguishing between similar models.

VI. Discussion and Future Work

The introduction of the CDRF benchmark and its accompanying toolkit represents a significant step toward addressing the critical need for robust, reproducible, and realistic evaluation of RF-based drone perception systems. By providing a large-scale dataset rich in device diversity and environmental conditions, coupled with a principled raw-signal augmentation pipeline, CDRF enables the research community to move beyond idealized benchmarks and tackle the challenges of real-world deployments. The strong performance of our baseline models in clean, controlled conditions underscores the quality of the collected data, while the performance degradation under augmentation highlights the existing gap between laboratory performance and operational robustness.

Our experiments reveal several key challenges and opportunities. The drop in recall for the YOLOv11n detector under augmented conditions suggests that while our whole-signal annotation policy is effective, more advanced architectures or training strategies may be needed to maintain high sensitivity in low-SNR and high-interference scenarios. Similarly, while our open-set recognition baseline shows promise, the task of reliably detecting unknown drone threats remains a significant challenge that requires further innovation in both model design and feature representation.

To ensure the long-term relevance and utility of CDRF, we are committed to its continuous expansion. Future work will focus on several key areas:

- **Dataset Expansion:** We will regularly update the dataset with new and emerging drone models, as well as more diverse background and interference signals captured from a wider range of environments (e.g., urban, industrial, and rural). We also plan to integrate CDRF with other public datasets to create an even more comprehensive and diverse benchmark.
- **Advanced Model Architectures:** We will explore the application of more sophisticated deep learning models, such as Transformers and graph neural networks, which may be better suited to capturing the complex temporal and spectral dependencies in RF signals.
- **Enhanced Augmentation Techniques:** We will continue to develop more advanced data augmentation techniques that can more realistically simulate a wider range of real-world channel effects, such as multipath fading and Doppler shifts.
- **Focus on Advanced Tasks:** We will further investigate the more challenging tasks of multi-label and open-set classification, which are critical for real-world applications where multiple drones may be present, and novel threats are a constant concern.

VII. Conclusion

In this paper, we introduced CDRF, a comprehensive benchmark dataset and open-source toolkit for RF-based drone detection, classification, and open-set recognition. CDRF addresses key limitations of existing datasets by providing a large-scale, diverse collection of real-world and synthetically augmented RF signals, coupled with a suite of tools for reproducible data processing, augmentation, and evaluation. Our extensive baseline experiments across a range of machine learning tasks demonstrate the utility of CDRF for developing and evaluating robust drone perception models. By open-sourcing the dataset and toolkit, we aim to foster a more collaborative and rigorous research environment, accelerating progress toward reliable and field-ready counter-drone systems. We believe that CDRF will serve as a valuable resource for the research community and a catalyst for innovation in the years to come.

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